



Human Computer Interaction Course COMP322

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Section:1

QOU Student Portal Redesign — Design Rationale:

1. Conceptual Model:

The new QOU Student Portal emphasizes instant information accessibility for the users. Rather than having to navigate several screens, all the vital information is now presented on one main screen, including:

- Daily Schedule
- Updated Grades
- Upcoming Tests/Assignments
- Course Information

The model prioritizes usability and reduces cognitive load, allowing students to complete tasks efficiently and with minimal navigation.

2. Persona-Based Design Considerations

Primary Persona: Mohammad Ahmad

Goals:

1. Check all grade publication statuses without opening individual course panels.
2. View weekly schedule without calculating odd/even weeks.

Frustrations addressed:

- Odd/Even week confusion
- Invisible individual assignment grades
- Unclear zero grades

Secondary Persona: Sujood

Goals:

1. Access grades, unread messages, and pending actions in under 3 minutes.
2. Quickly view and manage academic information relevant to her courses.

3. Complete tasks reliably within the portal without needing to visit the university.

Frustrations addressed:

Slow or unreliable administrative processes.

Messages hidden in submenus.

Unclear status of grades or exam reviews.

Some portal services appear to exist but are not available or functional when used.

3. Norman's Principles

The redesign of the QOU Student Portal applies Norman's six design principles to make the system easier to understand, faster to use, and less error-prone.

3.1 Affordances: The design of the interface provides the intended functions for the elements by their appearance and labeling. For instance, the labels on "View Grades", "Open Schedule", and "Read Message" buttons inform the student about what can be done through the buttons. It helps users understand what they can do without any additional explanations.

3.2 Signifiers: Clear visual and textual signifiers are used throughout the system to direct users to the next steps. For example, a red dot labeled "New" shows newly published grades or messages that have not been viewed yet. A greyed out button means that performing some actions such as reviewing exams is currently impossible.

3.3 Mapping: The design of the new portal corresponds to the student thinking style in respect to task completion. First of all, the most frequently used data appears on the portal dashboard: system status summary, schedule, grades, messages, course services and Exams, it allows answering the natural student question "What should I know today?" instead of making a student find required information in different sections of the portal

3.4 Feedback: Feedback provided by the portal is direct and appropriate. If no grade is available, the system displays "Not yet published" instead of leaving the field blank or showing zero. This prevents students from confusing an unpublished grade with a real grade of zero. Also, when an exam review is not available, the system clearly shows that the review period is closed instead of allowing the student to open a page that only displays an error message.

3.5 Constraints: Constraints are used in the redesigned portal to reduce mistakes and prevent students from misunderstanding schedule and grade information. In the weekly schedule, the system does not require students to calculate whether the current week is odd or even. Instead, it automatically displays the schedule for the current week and provides an option to view the schedule for any selected week. This keeps the odd/even logic inside the system rather than placing the burden on the student. In the grades section, the number 0 is displayed only when it is a real published grade. If the grade has not been released yet, the system shows “Not yet published.” These constraints help prevent wrong assumptions about classes, grades, and portal information.

3.6 Conceptual Model: It is essential to mention that the newly designed portal has the conceptual model: a dashboard where all necessary updates are gathered together. Users will be able to realize that their dashboard serves as the first tool that should be used to receive information about schedules, grades, messages, tests, assignments, and other useful material. Such a model is helpful for the project personas, as Mohammad needs immediate access to grades and schedules, whereas Sujood is to finish her portal task quickly.

4. Nielsen’s Heuristics

H1. Visibility of System Status: The revised portal structure makes sure that the students can observe the status of the most important subsystem all together from one summary card. Also they know the status of their grades or the exam review. There is a clear indication for each course grade whether it is “New grade published”, “Viewed” or “Not yet published”. Likewise, in case an exam review is not available, there is a clear indicator that the review process has been closed or not opened yet. Thus, no confusion is left to the students regarding the status of any grades or exam reviews.

H2. Match Between System and the Real World: In the redesigned portal, the user-friendly language is used instead of using technical terms and jargon. In particular, the headings on the schedule page will be something like “Classes for this week” instead of asking users to figure out whether it is an even or an odd week. The term “Not yet published” is used in place of blank cells in the case of the grades report.

H3. User Control and Freedom: The redesigned portal gives students clear control over their navigation without forcing them through unnecessary steps. From the dashboard, students can open the grades section, weekly schedule, exams, or assignments, then return easily to the main dashboard. The schedule view also allows students to check the current week or select another week when needed. This helps students move between important academic information without getting lost inside long menus or repeated pages.

H4. Consistency and Standards: Consistent layouts, wording, and interaction principles have been employed in the redesigned portal. The same card designs and statuses have been applied to grades, messages, exams, and assignments. There would be no need for students to wonder about the meaning of different words or visual styles, as defined by Nielsen.

H5. Error Prevention: Errors are avoided from the very beginning through the prevention approach. Links to unavailable exams for revision will be turned off and not activated; unpublishable grades are clearly marked and not represented as zeros. Additionally, the calendar does not imply the need for manual calculation of the odd/even weeks' numbers. This way, less errors regarding missed lessons, misunderstandings related to grades, and unnecessary visits to the faculty can occur.

H6. Recognition Rather than recall: With the portal's interface redesign, the load of users' memory is decreased. The students do not need to think what week number they are studying at now, whether it is odd or even. In addition, there is no need for them to recollect which subject might have some new message or grade.

H7. Flexibility and Efficiency of Use: The new dashboard provides adaptability and efficiency because it gives the ability to students to review the significant changes before accessing each section. The status indicators in the navigation menu indicate the current status of the grades and exams sections in red, green, or grey according to whether there is something new, viewed earlier, or nothing new in the section. In this way, they are aware of when there is something new, so they do not waste additional time accessing the same sections. Thus, both Mohammad and Sujood can access relevant information on their phones easily and within a few clicks.

H8. Aesthetic and minimalist design: Unlike the old portal that is characterized by too many unnecessary columns and information-filled tables, this redesigned interface utilizes simple cards and labels. The task-related information only is presented. For example, in the schedule tab, only the relevant columns are displayed such as class name, time, type, and location while also preserving the functionality of viewing more details that were moved to a pop-ups as a sort of layering information based on their relevance.

H9. Help users with errors: Since the redesigned portal focuses on preventing errors before they happen, this heuristic is addressed only when an error or unavailable action still occurs. For example, if a student tries to access an unavailable service, the system should display a clear message explaining that the service is currently not available, instead of showing a blank page or a technical error.

H10. Help and Documentation: Help information is provided if necessary in the form of concise tips. For instance, “Not yet published” could be further explained by a tooltip, as well as why a certain “exam review” button is greyed out. This meets the requirements of Nielsen who suggests that the help information be task-related, concise and easily accessible.

5. Cognitive Load Considerations

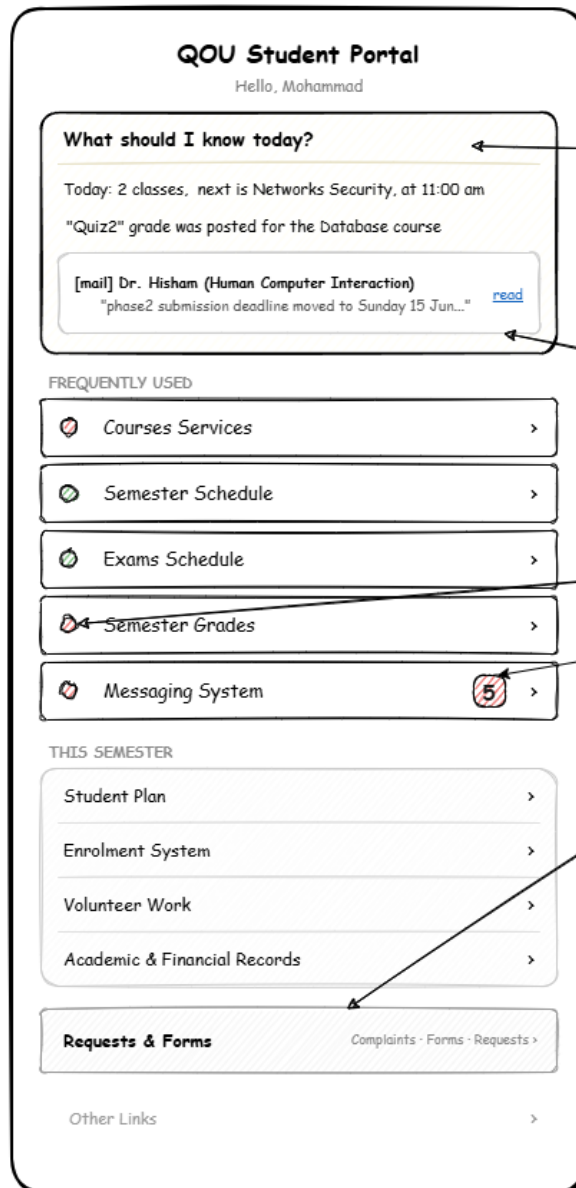
- Students do not need to calculate odd/even weeks manually because the system displays the current weekly schedule automatically.
- Students can check grade status from one overview screen without opening each course separately.
- Schedule and grade pages avoid displaying unnecessary information that does not help the student complete the task, least relevant information were moved to pop-ups.
- Cards, status indicators, and clear labels such as “New,” “Viewed,” and “Not yet published” reduce the mental effort needed to understand portal information.

6. Accessibility

- - Full keyboard navigation support.
- - Screen reader compatible text labels.
- - WCAG 2.2 compliant color contrast.
- -Status changes are not communicated by color alone. Each color indicator is paired with a clear text label and icon, such as “New,” “Viewed,” or “Not available,” so color-blind users can still understand the status.

Low-fidelity wireframes:

Dashboard Wireframe:-



dashboard: why was it built like this:

this screen Summarizes the conceptual model. the dashboard is the "first tool" a student opens, it answers "what should I know today?" at first glance instead of making them visit sections one by one. the summary card combines grade status, unread messages and today's next class the moment it opens, that is Sujood's Requirement 3 (everything she needs in under-3-min session). now it became a 20 seconds job.

Requirement 6: the latest faculty message is readable right here in the preview card, you tap "read" without entering the messages submenu. this kills Violation 7 (faculty messages buried two levels deep) and reaches Mohammad's Goal 3. the Messaging link still keeps its unread-count badge (5) so you also see how many are waiting.

H7 + status propagation: the frequently-used links each carry a three-state dot (red = new/unseen, green = seen, grey = nothing at all to see). this is exactly where the red dots from Semester Grades and Courses Services travel up to (by transitivity). you see "something is new in grades" from the dashboard before opening anything. there is no need for the 3 minutes check up to see if there is anything new. one look at the dashboard and you know the status of the system and your next move instantly.

The application of Layering(frequently used easy to access while rarely used can be two levels buried): (H8 minimalist design). most frequently used items get status dots and previews. while the 4 mid-use items ("This semester") are links grouped in one section. The 3 rarely-used services are gathered under one "Requests & Forms" link (here was the layering applied). And finally, "Other Links" sits at the far bottom. order never changes between visits Sujood navigates .by muscle memory (that is recognition rather than recall)

Semester Schedule Wireframe:

Semester Schedule

Current Week: Week 7
3 June - 9 June

Select Week

Print

The system shows the current week automatically. Students do not calculate Odd/Even weeks.

Course	Day	Time	Type	Location
Electrical Circuits	Mon	12:00-13:00	Campus	Computer Lab 7
Computer Logic	Tue	11:00-12:30	Campus	Room 304
Operations Research	Sat	16:00-17:30	Online	Join link
Algorithms	—	No meeting time	Not scheduled	—

Course Details Pop-up

Electrical Circuits | Instructor: Dr. Abdullah
Section: 1 | Credit Hours: 3
Extra details are hidden here instead of crowding the table

Legend: Grey row = no scheduled meeting | Join link = online class | Details appear in pop-up

Annotation: This semester schedule redesign solves the main problems found in Phase 1. It removes unnecessary columns, shows the current week automatically, avoids Odd/Even week calculations, uses clear Online/Campus labels, and explains missing meeting times instead of leaving empty cells.

Design Rationale

1. Dense old table → simplified columns only
2. Odd/Even confusion → current week shown automatically
3. Online hall inconsistency → consistent Online + Join link
4. Empty fields → clear message: No meeting time
5. Extra course details → moved to pop-up

Select weeks drop down:

Select Week ▾

Current week

Next week

Or choose a date

DD / MM / YYYY

Course Details pop up Wireframe:

Course Details

Course Name

Electrical Circuits

Instructor

Dr. Abdullah

Section

1

Credit Hours

3

Note

Extra details are hidden here instead of crowding the table

Close

Exam Schedule Wireframe:

Exam Schedule

Current Semester
Second Semester 2025/2026

Filter

Select Semester

Upcoming Exams: 6
Next Exam: 16 May — Networks
Exam Review: 1 available, 2 closed

Upcoming exams are shown automatically. Filtering is optional, not required before viewing.

Course	Date	Time	Type	Location	Review
Networks	16-05	11:00-12:30	Final Exam	Room 305	Closed
Computer Logic	17-05	11:00-12:30	Final Exam	Room 102	Available
Systems Analysis	18-05	13:00-14:30	Final Exam	Room 402	Not yet

No other exams scheduled this week

Exam Details Pop-up

Computer Logic | Final Exam
Date: 17-05-2026 | Time: 11:00-12:30
Building: Hassan Al-Harazeen | Room: 102 | Seat: 10

Annotation: This exam schedule redesign solves the main Phase 1 problems. Upcoming exams are shown immediately without forcing the student to choose exam type first. The exam labels are simplified, irrelevant columns are removed, and exam review availability is shown clearly as Available, Closed, or Not yet available.

Key Design Decisions

1. Exams shown by default → no required pre-filtering

2. Confusing exam types → simple labels like Final Exam

3. Dense exam table → only course, date, time, type, location, review

4. Review icon confusion → clear status: Available / Closed / Not yet

5. Extra details → moved to exam details pop-up

Filtering Wireframe:

9 | Page

Exam Schedule — Filter

Filter options

Exam type

☒ Midterm

☒ Final Exam

☐ Incomplete

Filter by date

From

DD / MM / YYYY

To

DD / MM / YYYY

Course name

Search course...

All three filters can work together

Apply filters

Clear all

Showing 2 results

Course	Date	Time	Type	Location	Review
Computer Logic	17-05	11:00 - 12:30	Final Exam	Room 102	Available
Networks	16-05	11:00 - 12:30	Final Exam	Room 305	Closed

Screen reader note

This filter is especially useful for users with visual impairments using screen readers — all filters have proper labels, checkboxes are grouped with aria-label, and results table has a caption so screen readers announce the context correctly.

Exam Details pop-up Wireframe:

Course

Computer Logic

Final Exam

Date

17 · 05 · 2026

Time

11:00 - 12:30

Building

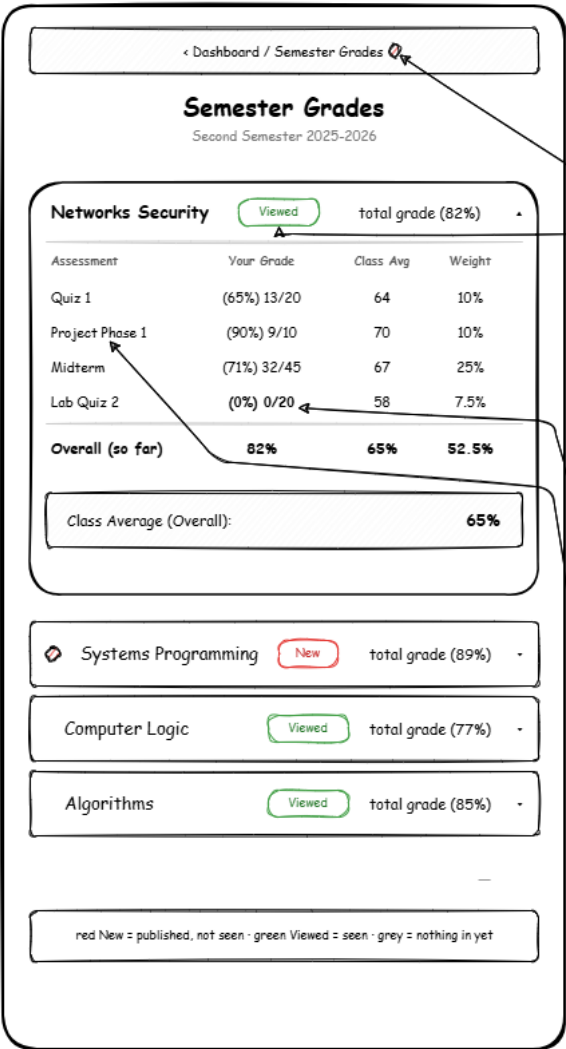
Hassan Al-Harazeen

Room

102

Close

Semester Grades Wireframe:



semester grades: why was it built like this:

Violation 13 in the old portal was that you hit "show grades" on every course one at a time and expand the courses one at a time just to know if anything was posted. now every course's status sits on one screen (H1). you tap a course only when you want the detailed breakdown and it expands right here in the same page (H1, H7). Also, status propagation (transitivity) was applied meaning that when a new grade gets published for a course the system pushes a red dot up to the "Semester Grades" entry in the main nav bar, and up to the dashboard. so you spot "something's new in grades" before you even open this screen. same rule we used on Course Services.

open = seen. the moment you expand a course that had a newly published grade its status changes from New(red) to Viewed(green) and its status propagates up. an open course never stays "new" - it is the same logic as the active tab on Course Services. that is consistency.

Scenario1 is one of the main reasons this screen matters. mohammad was frustrated by a row of 0s he couldn't read: real zero, or just not posted yet? look at Networks Security - Lab Quiz 2 shows "0/20" (a real published zero, it's a number) while "Not yet published" grades will not show up on this page at all. since assessments are now dynamic rather than static. so a present number can only mean a real grade. Requirement 5 fulfilled, and the exact thing that broke him in Scenario1 is gone.

Violation 14: no Course ID column. students know courses by name (H8). Violation 15: assessment rows use this course's own labels not one generic set forced on every course, so you can tell which mark is which (H2, H10). the breakdown carries the same columns as the Grades tab on Course Services - Your Grade, Class Average, Weight - so a student sees the same shape of data in both places (H4 consistency).

Course Services (Default appearance opened on the Exams service):

Course Services

Select Course Electrical Circuits ▼

i

 Course details[!] >

Exams ●

Grades

Links

Assignments

Exams

Exam Name	Date	Time	Place
Midterm Exam	10 Jun 2026	10:00 AM	Room 101
Final Exam	25 Jun 2026	02:00 PM	Main Hall
Practical Exam	Not Scheduled Yet	Not Scheduled Yet	TBD

1) Dropdown instead of multiple course sections to keep the same look and reduce clutter.

2) Button that clearly indicates it will show the course details when clicked.

3) Tabs for easy navigation between services.

4) Three-state indicator (● New, ● Viewed, ○ No Upcoming Exams) allows students to know if there are new exams without opening the panel.

Course Services (Assignments service):

Courses Services

Courses Services

Select Course Electrical Circuits

Course details [1]

Exams Grades Links **Assignments**

Assignments

Name · Due · Weight · Status

Assignment 2 - Circuit Analysis	Due 14 Jun · 15%	Not submitted
Lab Report 3	Due 18 Jun · 10%	Not submitted
Assignment 1 - Resistor Networks	Due 28 May · 15%	Submitted
Quiz Prep Worksheet	Due 20 May · 5%	Submitted
Homework 1 - KCL & KVL	Due 6 May · 10%	Graded
Intro Participation Task	Due 22 Apr · 5%	Graded

what was enhanced in that design:

Violation8 (3/4): before, you had to open each course's card and assignment panel one by one just to see if an assignment got published. mohammad ended up asking the WhatsApp group if anything was published. this tab is where "upcoming assignments" finally shows up in one place. and if any assignment got published it's "course name" in the drop down would have a red dot next to it's name, and by transitivity the "courses services" label in the main navigation bar would have a red dot identifying something requiring attention.

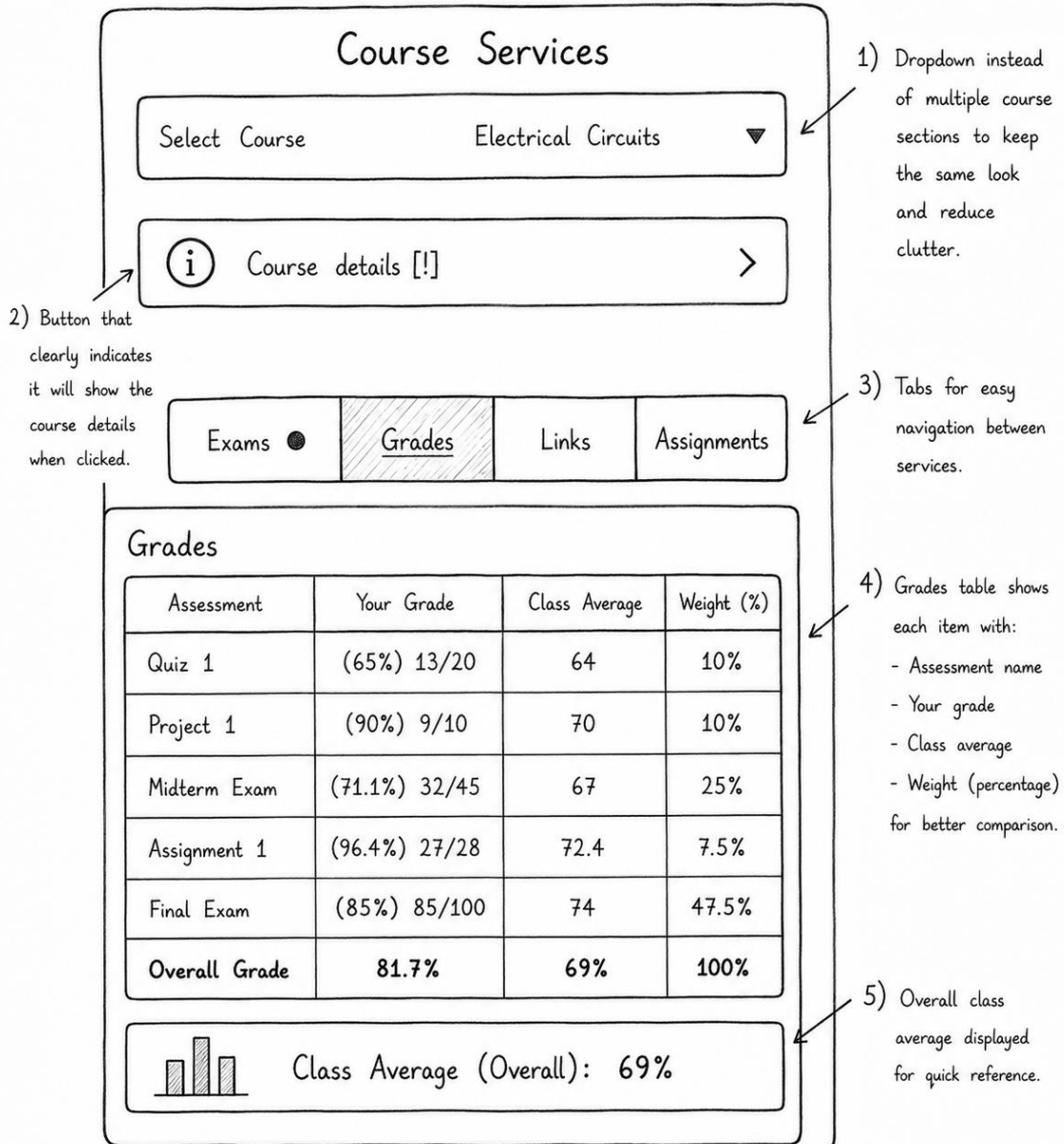
1) one dropdown, not a section per course (V5). same shell as Exams/Grades so it all matches: H4 consistency and H8 minimalist design.

2) tab strip carries the 3-state dot (red=new, green=seen, grey=empty). the V8 fix. Assign. tab is active here so it doesn't have a red dot. while the exams has a red dot.

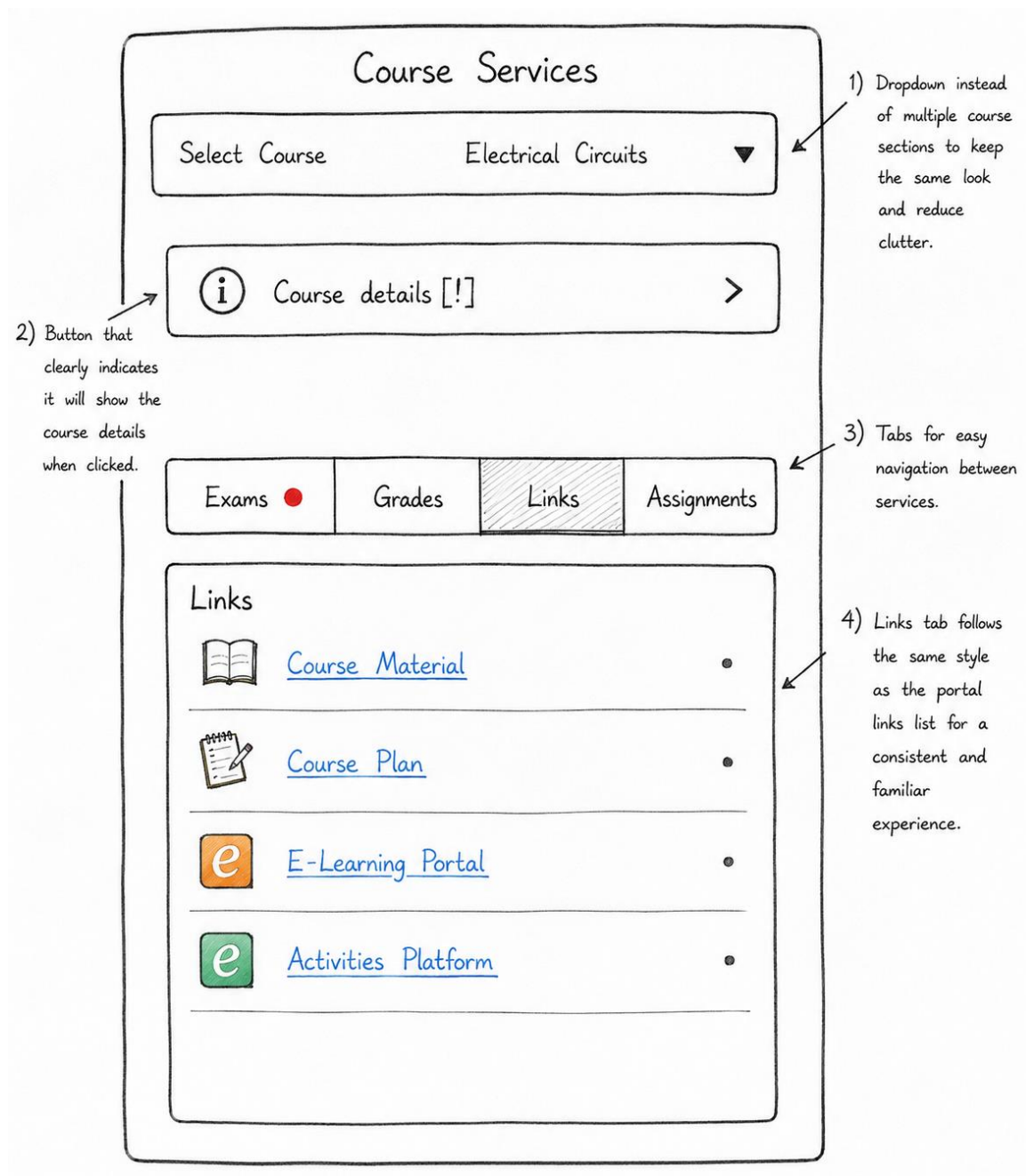
3) each row = name + due + weight + status. weight matters because it tells you what the thing is actually worth (ties to V15, the old static unlabelled assessments). sorted Not submitted → Submitted → Graded and the not submitted is sorted by date so the stuff needing action sits on top. fits Mohammad's open-do-one-thing-leave pattern.

4) status chip not a fake number. no "0" that could mean "not marked yet" (R5 / S1). Following the mapping principal or the H2 of nielsen's heuristics

Course Services (Grades Service):

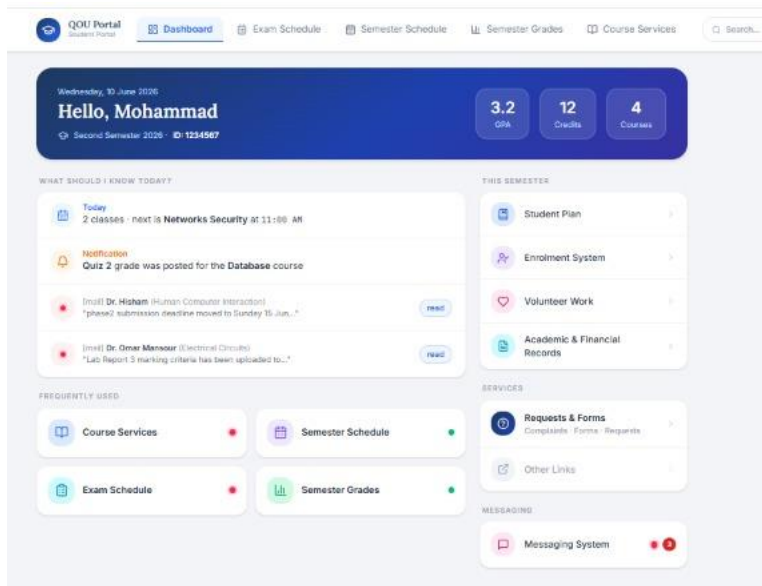


Course Services (Links Service):

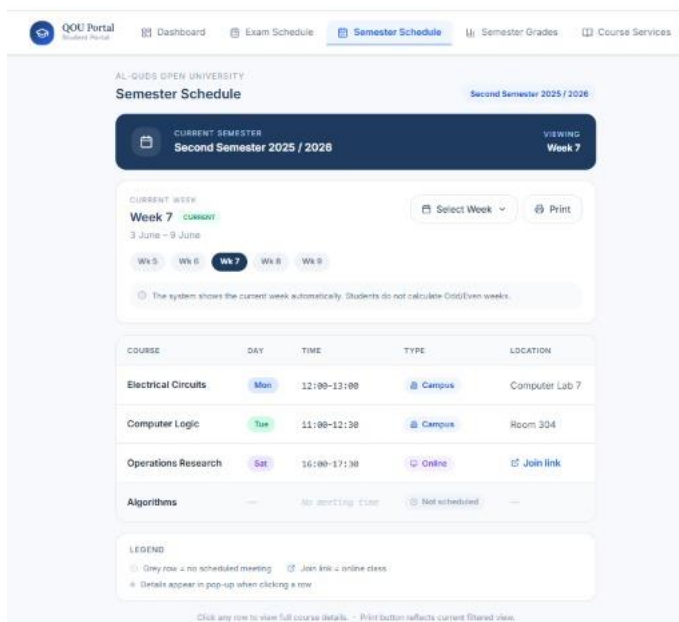


Interactive prototype:

Dashboard Wireframe:-



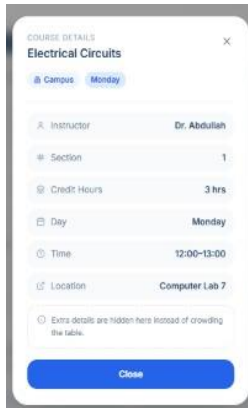
Semester Schedule Wireframe:



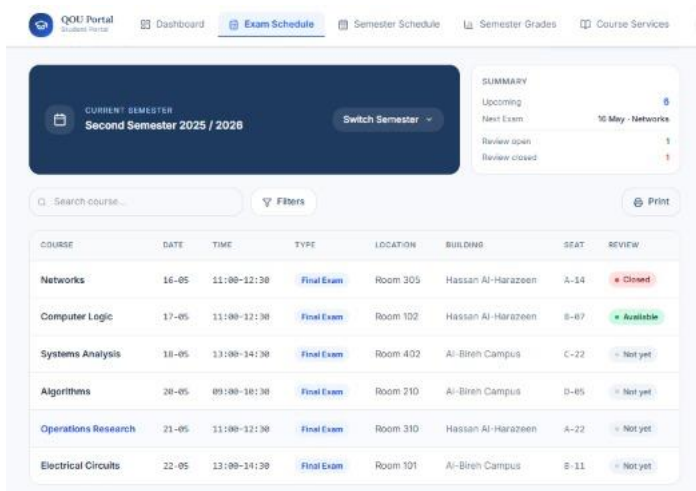
Select weeks drop down:



Course Details pop up Wireframe:



Exam Schedule Wireframe:



Filtering Wireframe:

Filter Exams

COURSE NAME

Search...

EXAM TYPE

☐ Final Exam

☐ Midterm

☐ Practical

REVIEW STATUS

Available

Closed

Not yet

DATE RANGE

From

mm/dd/yyyy

To

mm/dd/yyyy

PREVIEW - 6 RESULTS

Networks	16-05 - 11:00-12:30	Closed
Computer Logic	17-05 - 11:00-12:30	Available
Systems Analysis	18-05 - 12:00-14:30	Not yet
Algorithms	20-05 - 09:00-10:30	Not yet
Operations Research	21-05 - 11:00-12:30	Not yet
Electrical Circuits	02-06 - 10:00-14:30	Not yet

Clear All

Apply

Exam Details pop-up Wireframe:

EXAM DETAILS

Computer Logic

Final Exam

Review Available

Date

17-05

Time

11:00-12:30

Location

Room 102

Building

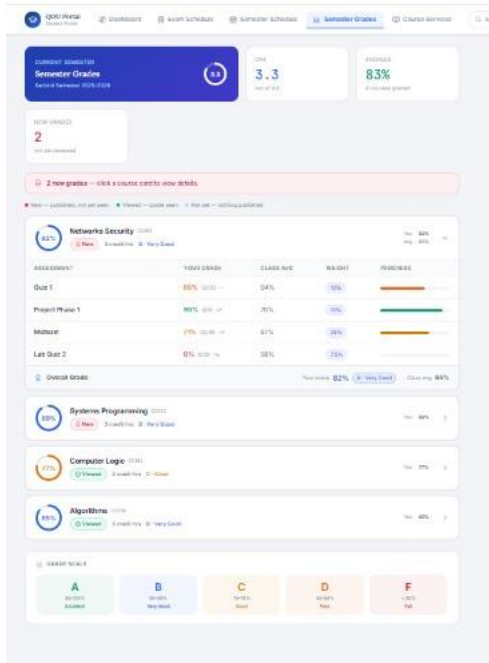
Hassan Al-Harazeen

Seat

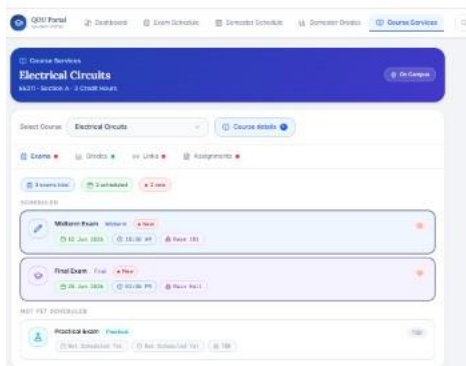
B-07

Close

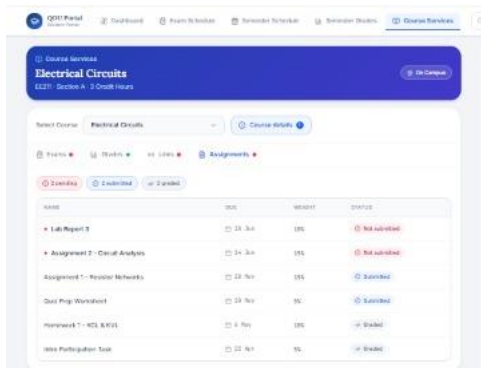
Semester Grades Wireframe:



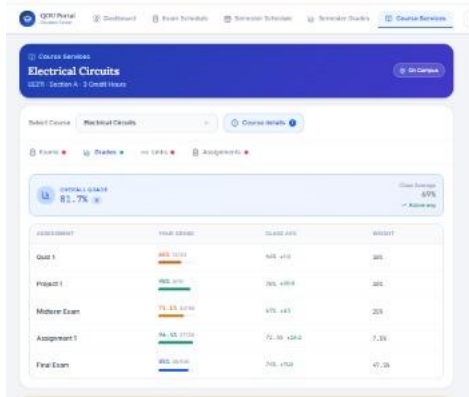
Course Services (Default appearance opened on the Exams service):



Course Services (Assignments service):



Course Services (Grades Service):



Course Services
Electrical Circuits
0229 Section A - 3 Credit Hours

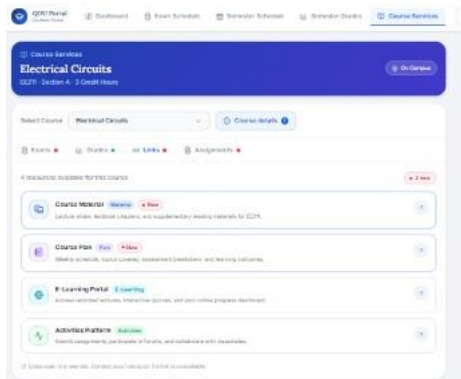
Select Course: **Electrical Circuits** [Course Details](#)

Exams **Grades** **100.00%** **Assignments**

OVERALL GRADE
81.7% [View Average](#) [Report Avg](#)

ASSIGNMENT	YOUR GRADE	CLASS AVG	WISDOM
Quiz 1	85% (10/12)	74% (9/12)	20%
Project 1	90% (9/10)	70% (8/10)	20%
Midterm Exam	79.1% (32/40)	67% (41)	20%
Assignment 1	76.3% (17/22)	72.3% (32/44)	7.3%
Final Exam	85% (26/30)	74% (25/34)	47.3%

Course Services (Links Service):



Course Services
Electrical Circuits
0229 Section A - 3 Credit Hours

Select Course: **Electrical Circuits** [Course Details](#)

Exams **Grades** **100.00%** **Assignments** **Links**

4 RESOURCES AVAILABLE FOR THIS COURSE [+ 2 more](#)

Course Materials View + New	Lecture notes, textbook chapters, and supplementary reading materials for 0229.
Course Plan View + New	Weekly schedule, study calendar, assignment deadlines, and upcoming activities.
e-Learning Portal e-Learning	Access virtual activities, interactive lessons, and your online program dashboard.
Activities Platform Activities	Engage with peers, participate in forums, and collaborate with classmates.

If you need more resources, contact your instructor for additional recommendations.